

Toward Lightweight Aerial 5G Cells: A Reproducible Real-Radio OAI Split-DU Testbed on Commodity Arm Platforms

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Abstract—For a battery-powered UAV cell, every gram and watt devoted to RAN compute competes with flight-duration margin. We therefore keep the 5G Core (5GC) and central unit (CU) on the ground and design the airborne unit around only the backhaul endpoint, distributed unit (DU), and radio. The key question is then practical: can widely available commodity Arm computers sustain a real-radio OpenAirInterface (OAI) 5G DU with useful performance over heterogeneous F1? We answer it with Raspberry Pi 5 and Jetson Orin Nano DUs, a USRP B210, a commercial handset, and Ethernet, Wi-Fi/GRE, and 5G/WireGuard backhaul. Twenty DL and 20 UL trials across ten configurations yield 400 throughput observations. Jetson preserves 88.4% of x86 split-DU Ethernet DL throughput and 87.5–89.0% across all three bearers; 5G/WireGuard preserves 76.4–77.1% of each host’s wired DL rate. The validated Jetson/B210/RM500Q-GL electronics weigh 657.4 g and draw about 28 W under sustained traffic (757.4 g with integration allowance). A controlled link-adaptation intervention raises split DL from 23.4 to 99.4 Mb/s as dominant MCS moves from 3 to 26. Finally, synchronized radio, F1-U, CPU, and UHD evidence narrows the remaining monolithic-split gap to split-path scheduling/timing behavior. Beyond performance, this capability serves the emergency-response use case that motivates the aerial cell: once deployed above an affected area, it can broadcast a PWS warning to compatible handsets in the cell without knowing or addressing individual phone numbers. We release the OAI patch that carries the single-segment Write-Replace Warning over F1 from CU to DU, where SIB8 is scheduled to the handset; to the best of our knowledge, this is the first publicly documented OAI PWS-over-F1 path. The public artifact makes the payload, performance, and emergency-broadcast baseline reproducible with laboratory-accessible hardware.

Index Terms—5G NR, OpenAirInterface, CU/DU split, F1, AArch64, Arm, testbed, software-defined radio, wireless backhaul, UAV, aerial RAN, public warning system

I. INTRODUCTION

UAV-mounted cellular cells are attractive for temporary coverage, emergency response, and rapidly deployable networks, but they face a hard systems constraint: payload mass and electrical load directly consume the aircraft’s limited energy budget. Carrying a complete base station therefore spends valuable airborne resources on functions that do not need to be close to the radio. The 3GPP F1 split provides a natural boundary: RRC/PDCP/SDAP remain in the CU on the ground, while the latency-sensitive RLC/MAC/PHY stay in the DU next to the radio; F1-C uses F1AP/SCTP and F1-U uses GTP-U/UDP [1], [2]. The intended airborne payload

can consequently be reduced to a backhaul endpoint, DU, SDR, and antennas rather than an entire 5GC/gNB stack. In a disaster area, a particularly valuable first service from such a temporary cell is an emergency broadcast: PWS can deliver the same warning or instruction to compatible handsets in the covered area without a list of recipient phone numbers. This makes PWS a natural service requirement for the split aerial architecture, not an unrelated add-on.

This architecture only helps if the airborne DU is small and practical without sacrificing too much radio performance. We deliberately evaluate two off-the-shelf Arm platforms—Raspberry Pi 5 and Jetson Orin Nano—rather than custom flight computers. They are widely available, well documented, and easy for academic laboratories to source, while spanning a useful range from a low-complexity single-board computer to a higher-performance embedded platform. The objective is not to argue that Arm is intrinsically superior to x86; it is to quantify the performance price paid when choosing commodity embedded compute to make a reproducible airborne DU practical.

This leads to three experimental questions. *RQ1*: how much throughput is lost when the DU moves from x86 to commodity Arm under the same real-radio access configuration? *RQ2*: how much additional penalty appears when F1 moves from wired Ethernet to untethered Wi-Fi/GRE or 5G/WireGuard? *RQ3*: which lower-layer failures can destroy useful throughput even while F1, registration, and IP connectivity remain healthy? We answer them with a fixed B210/UE/n78 access profile, repeated measurements, and a layered acceptance contract.

The paper answers these questions with three measured results and contributes one additional split-RAN capability:

- a reproducible UAV-oriented remote-cell design that keeps 5GC/CU on the ground, uses laboratory-accessible Arm DUs, and measures the integrated Jetson/B210/5G-backhaul payload at 657.4 g and approximately 28 W;
- a 400-observation host-bearer benchmark that quantifies the throughput cost of embedded compute and heterogeneous F1 under a fixed 106-PRB real-radio configuration;
- an instrumented failure study: a controlled BLER/MCS intervention recovers a $4.25\times$ throughput collapse, while synchronized mono/split telemetry narrows the residual split penalty to scheduling/timing behavior; and

- to the best of our knowledge, the first publicly documented OAI PWS-over-F1 implementation: our released patch carries the F1AP Write-Replace Warning procedure from the ground CU to the remote DU, enabling area-wide emergency broadcast to compatible handsets without per-phone-number addressing [3].

Concurrent two-DU operation is retained as a first-step replication check. The novelty is therefore a measured and reproducible *airborne payload/performance trade-off* plus an enabling public-service function for the split architecture, not the aerial-DU or wireless-F1 concept by itself.

II. RELATED WORK AND POSITIONING

Aerial cellular prototypes established the motivation for moving RAN functions onto aircraft. Flying Rebots, SkyRAN, and SkyCell demonstrated that open SDR cellular stacks can operate aloft on commodity x86-class compute [4], [5], [6]. Their key cost, from our perspective, is that substantial base-station compute travels with every aircraft. An F1 split changes that scaling rule by sharing CU/5GC functions on the ground and repeating only the remote DU/radio payload.

The closest predecessor is Mundlamuri *et al.*, who demonstrated an OAI 5G aerial DU with a B200mini, Quectel 5G modem, wireless F1, and commercial terminals [7]. We therefore do not claim the aerial-DU concept or wireless F1 as novel. Our question is different: what happens when the airborne DU is deliberately built from widely reproducible Arm computers, and how much performance is traded for that payload choice across several F1 paths?

Recent non-aerial OAI systems provide complementary evidence. Bui and Tran implement an OAI CU/DU MU-MIMO testbed [8], while Beurdouche and Knopp study intensive vRAN deployment across server-class compute architectures [9]. OAI itself provides AArch64 portability through SIMDe [10], and a 2024 6GSPACE report documents Raspberry Pi execution with the RF simulator [11]. These works show that compute matters, but do not provide the same real-radio, commercial-UE host-backhaul comparison on commodity Arm. Prior PWS work studies warning-system security [12]. Abou Hasna *et al.* recently added open-source SIB8 transmission to a monolithic OAI gNB and kept the implementation modular specifically to ease future CU/DU support [13]. We close that disaggregation gap: our published patch implements the F1AP Write-Replace Warning path from CU to DU and validates the resulting SIB8 on a commercial handset [3]. To the best of our knowledge, no earlier public OAI implementation demonstrates PWS delivery through an actual F1 CU/DU split.

III. TESTBED AND METHOD

A. UAV-Oriented Architecture and Commodity Arm Choice

Fig. 1 makes the deployment boundary explicit. The x86 ground host keeps the OAI 5GC/CU; only the backhaul endpoint, Arm DU, B210, and antennas form the intended airborne unit. The DU runs RLC/MAC/PHY and drives the access B210 over USB 3. The access cell is fixed at band n78,

3.6192 GHz, 30-kHz SCS, and 106 PRBs, with a commercial Nothing Phone handset as the end-to-end UE. Ethernet provides the deterministic/tethered reference, while Wi-Fi/GRE and 5G/WireGuard represent untethered F1 options. In the 5G case, an RM500Q-GL modem camps on an independent donor cell so the access cell never backhauls itself [14].

Raspberry Pi 5 and Jetson Orin Nano are chosen intentionally as reproducible lab platforms rather than bespoke avionics. Pi 5 provides a widely available low-complexity Arm baseline; Jetson represents the higher-performance embedded choice. The experiment therefore measures a design trade-off a laboratory can actually reproduce: how much radio performance is retained when the x86 DU is replaced by hardware that is realistic to carry and easy to source.

The three F1 modes are implemented as distinct, packet-verifiable paths rather than aliases over the same management network. The Ethernet profile uses a 9000-byte MTU on the CU/DU path and corresponding core bridge/veth path. The Wi-Fi profile carries the F1 addresses inside GRE over the wireless underlay. For 5G/WireGuard, one B210 serves the access UE and a second, independent donor B210 serves the RM500Q-GL; F1 uses the WireGuard inner interface while packet capture also verifies the modem-facing outer path. Every run checks that F1-C and F1-U do not leak to an unselected Ethernet, Wi-Fi, or management route [14]. This matters experimentally because a healthy SCTP association on the wrong interface would otherwise invalidate any backhaul comparison.

The source tree is frozen at commit 102965a6 and built natively on Arm through OAI's SIMDe-backed AArch64 path [10]. Jetson uses a matching L4T R36.4.4 / JetPack 6.2 SCTP-enabled kernel and rollback entry [15]. Stable B210 operation requires SuperSpeed USB, locked clocks, disabled USB autosuspend, bounded DU CPU placement, and xHCI interrupt isolation; USB 2 produces continuous UHD overflows. These details are retained because they determine whether the "lightweight" platform is a usable real-radio DU, while the full kernel procedure remains in the artifact.

B. Experimental Control and Measurement

A run is accepted only if five independent gates pass: (i) core health; (ii) F1-C/SCTP and F1-U/GTP-U establishment; (iii) packet capture proving that F1 uses the selected bearer rather than a management interface; (iv) UE registration, PDU session, external reachability, and PWS reception; and (v) clean teardown and Ethernet rollback. The isolated test PLMN operates inside a Faraday cage with test-only warning text. Host/radio identity, routes, USB speed, OAI revision, and patch state are recorded before each run [14].

Each host-bearer configuration uses 20 repetitions with the same UE, B210, n78/106-PRB profile, OAI revision, and gate sequence. iperf3 TCP uses one stream for 30 s after a 5 s warm-up; DL and UL are measured separately, scenario order is randomized, and runs are separated by 10 s. The server is on the 5GC/CU side, so throughput traverses the complete UE-gNB-5GC user plane. Reported repeated values

TABLE I
POSITIONING AGAINST REPRESENTATIVE AERIAL AND OAI SPLIT-RAN SYSTEMS. VALUES ACROSS PAPERS ARE NOT DIRECT PERFORMANCE RANKINGS BECAUSE RADIOS, BANDWIDTHS, UES, AND TRAFFIC DIFFER.

System	RAN / Compute	Experimental focus	Gap Relative to This Work
Flying Robots [4]	OAI LTE eNB; x86; USRP	Aerial cellular relay	Complete eNB aloft; no 5G F1 split or embedded Arm DU.
Mundlamuri <i>et al.</i> [7]	OAI 5G DU; B200mini; Quectel wireless F1	Aerial DU; ~ 30 Mb/s to commercial terminals	Closest remote-DU predecessor; no repeated Arm host-bearer matrix or failure diagnosis.
Bui and Tran [8]	OAI 5GC/CU/DU; multi-UE NR	MU-MIMO implementation and throughput	Functional CU/DU testbed; not embedded Arm or heterogeneous F1 characterization.
Beurdouche and Knopp [9]	OAI vRAN; server-class GPP/accelerator platforms	Intensive multi-instance vRAN compute scaling	Compute-focused central deployment; not remote Arm DUs with B210/UE and variable F1.
Abou Hasna <i>et al.</i> [13]	Modified monolithic OAI gNB; B210	Open-source SIB8/PWS security testbed	Public SIB8 broadcast, but CU/DU disaggregation left as future support.
This work	Ground 5GC/CU; commodity Arm remote DU; B210; three F1 paths	400 repeated DL/UL observations; PWS over F1; payload power/mass; failure diagnosis	Reproducible UAV-oriented study plus published CU-to-DU PWS Write-Replace Warning patch.

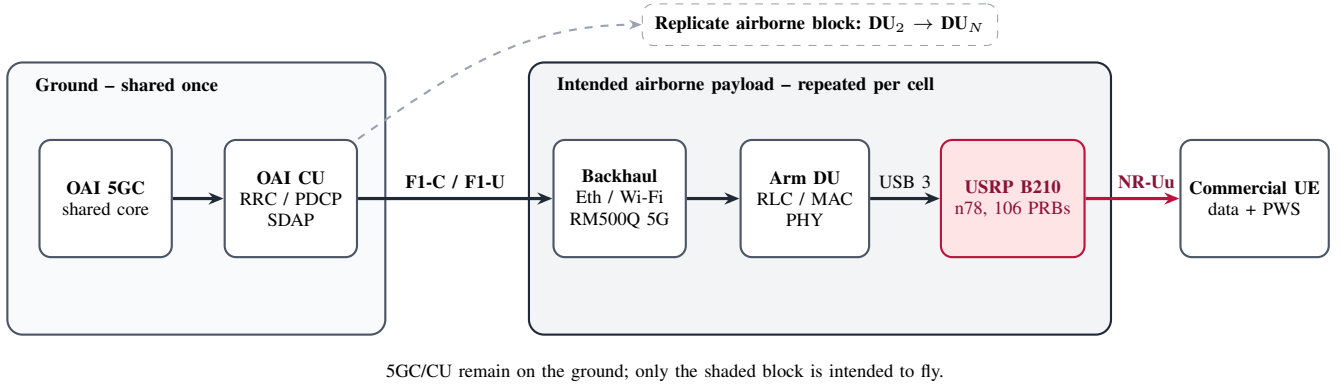


Fig. 1. UAV-oriented split architecture. Shared 5GC/CU compute remains on the ground, while the repeated airborne unit contains only the F1 backhaul endpoint, commodity Arm DU, and access radio. The solid path is the measured single-DU configuration; concurrent DU_2 is a supporting check and larger fan-out is future work.

TABLE II
EXPERIMENTAL CONFIGURATION HELD FIXED UNLESS VARIED

Item	Configuration
Deployment split	Ground: x86 5GC/CU; intended airborne unit: backhaul + DU + B210
OAI	commit 102965a6
Access radio	USRP B210, USB 3; n78, 3.6192 GHz, 30-kHz SCS, 106 PRBs
DU hosts	x86 reference; Raspberry Pi 5 (16 GB); Jetson Orin Nano
UE	commercial Nothing Phone handset
F1 paths	Ethernet (MTU 9000); Wi-Fi/GRE; RM500Q-GL 5G/WireGuard
Jetson SW	L4T R36.4.4 / JetPack 6.2; SCTP-enabled kernel; locked clocks
Traffic	iperf3 TCP, one stream, 30 s + 5 s warm-up, separate DL/UL
RF control	Faraday cage; same B210/UE/access profile for paired tests

are arithmetic means over the 20 valid repetitions; failed service gates invalidate a run rather than create a zero-rate sample.

Absolute RTT is not treated as a bearer metric because it depends on responder placement and ground-network routing. Delay is instead reported as the paired location-induced change $\Delta RTT(\ell) = RTT(\ell) - RTT(\ell_0)$, where ℓ_0 is the on-premises CU-side responder and ℓ is the same service at a more distant location; bearer, load, packet size, and test host are held fixed.

Table III separates transport context from handset-visible end-to-end throughput. The untethered paths impose clear throughput penalties, but delay is not inferred from their raw RTT because that number depends on endpoint placement and route selection. Delay is therefore reported through location-

TABLE III
MEASURED x86 SPLIT F1-BEARER THROUGHPUT BASELINE ($n = 20$)

F1 bearer	DL throughput (Mb/s)	UL throughput (Mb/s)
Ethernet	99.4	22.6
Wi-Fi/GRE	51.8	12.1
5G/WireGuard	76.2	18.9

normalized ΔRTT , while bearer substitution remains an end-to-end systems variable including contention, encapsulation, host networking, and the modem/underlay.

IV. RESULTS

A. The Cost of Moving the DU from x86 to Commodity Arm

Fig. 2 turns the airborne-compute choice into a measured trade-off. On Ethernet, x86, Jetson, and Pi 5 split DUs reach 99.4/22.6, 87.9/18.1, and 61.8/15.4 Mb/s DL/UL. Jetson therefore preserves 88.4% of x86 DL while moving the DU onto an embedded board; Pi 5 preserves 62.2%. More importantly, the Jetson/x86 ratio remains 87.5–89.0% across all three bearers. Among the tested embedded candidates, Jetson is therefore the strongest performance-preserving choice for the intended airborne DU, while Pi 5 provides a simpler reproducible baseline at a larger throughput cost.

The host ranking is the same across Ethernet, Wi-Fi/GRE, and 5G/WireGuard, showing that the Arm penalty is not a single-bearer accident.

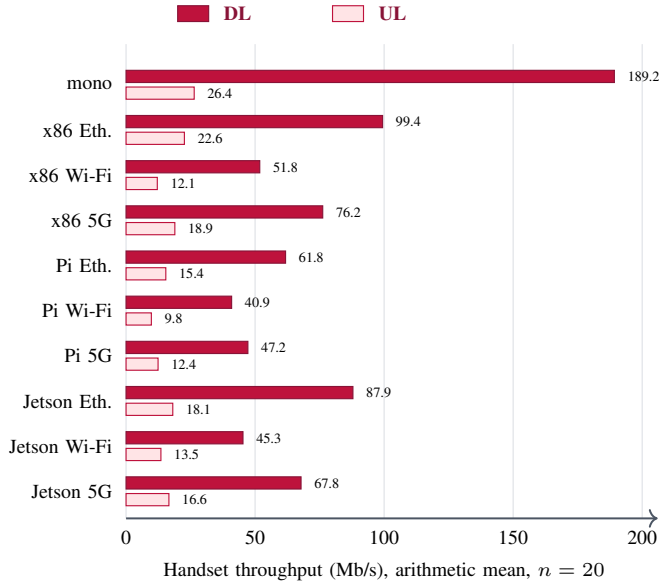


Fig. 2. Handset DL/UL throughput across ten configurations (arithmetic mean, $n = 20$ per setup; 400 observations).

B. The Cost of Untethering F1

Normalizing each host to its own Ethernet result isolates the transport side of the design. 5G/WireGuard retains 76.7% (x86), 77.1% (Jetson), and 76.4% (Pi 5) of wired DL; Wi-Fi/GRE retains 52.1%, 51.5%, and 66.2%, respectively. Thus the 5G/WireGuard penalty is remarkably consistent across very different DU hosts in this setup. It should not be read as a general 5G versus Wi-Fi ranking: the Wi-Fi route is contended, whereas the RM500Q-GL uses an independent donor cell. Rather, it quantifies the cost of replacing the tethered reference by the actual untethered paths available to this testbed.

UL follows the same broad host ordering but in a narrower dynamic range. This direction asymmetry matters later: x86 monolithic-to-split DL drops 47.5%, whereas UL drops only 14.4%. Treating “the split penalty” as one scalar would hide that scheduler/traffic-direction effect.

C. A Hidden Failure: Link Adaptation Collapses While F1 Looks Healthy

In the initial x86 Ethernet split state, F1 and UE attachment remain healthy but mean DL is 23.4 Mb/s. OAI tracks an exponentially filtered BLER,

$$\hat{b}_k = 0.9\hat{b}_{k-1} + 0.1b_k, \quad (1)$$

and adapts MCS against configured lower/upper bounds [16]. With the default DL window 0.05–0.15, mean measured BLER is 0.22, above the upper bound, and dominant MCS collapses to 3 (2–4).

After observing this state, we deliberately choose 0.25–0.35 as a *diagnostic* DL window that brackets the measured operating BLER; it is not the result of an optimization sweep and is not proposed as a universal target. Changing only the BLER thresholds (DL 0.25–0.35, UL 0.15–0.35) raises dominant MCS to 26 (25–27), mean measured BLER to 0.28, and mean DL to 99.4 Mb/s, a $4.25\times$ gain. Radio attenuation,

TABLE IV
REPEATED BLER-THRESHOLD INTERVENTION ($n = 20$)

Metric	Default	Diagnostic retune
DL target window	0.05–0.15	0.25–0.35
DL throughput (Mb/s)	23.4	99.4
Filtered DL BLER	0.22	0.28
Dominant MCS (range)	3 (2–4)	26 (25–27)

TABLE V
X86 MONOLITHIC–ETHERNET SPLIT DIAGNOSTIC EVIDENCE

Metric	Monolithic	Split Ethernet
DL throughput (Mb/s)	189.2	99.4
UL throughput (Mb/s)	26.4	22.6
Dominant DL MCS	26	26
Filtered DL BLER	0.27	0.28
F1-U rate at 99.4 Mb/s app.	—	104.8 Mb/s
F1-U loss / persistent queue	—	0 / none
Busiest DU RT thread	—	62%
UHD O/L/U counters	0/0/0	0/0/0
PRB/TB observation	full/continuous	intermittent gaps / smaller TBs

BWP, TDD profile, UE, F1 bearer, and traffic generator are unchanged.

The higher post-retune BLER is therefore not a regression: it lies inside the relaxed diagnostic window and is paired with a much higher MCS. This before/after experiment shows why F1 setup and packet flow alone are insufficient health criteria, while avoiding the stronger claim that 0.25–0.35 is optimal for other channels or UEs.

D. Residual Monolithic–Split Gap

The x86 monolithic reference reaches mean DL of 189.2 Mb/s versus 99.4 Mb/s for the *retuned* Ethernet split, a 47.5% reduction; mean UL falls only 14.4% (26.4 to 22.6 Mb/s). We compare against the repaired split state specifically so the default BLER collapse is not mistaken for the explanation of this residual gap.

The measured radio operating point is nearly identical (MCS 26; BLER 0.27/0.28). Split F1-U carries 104.8 Mb/s, 5.4% above application throughput, with no observed loss or persistent queue; the busiest DU real-time thread is 62%, and UHD reports no O/L/U events. RF-quality collapse, Ethernet capacity, persistent F1-U backlog, and sustained gross CPU saturation are therefore poor primary explanations.

The remaining observed discriminator is scheduling occupancy: monolithic execution allocates the full 106-PRB budget on each active DL slot, whereas split execution shows inter-slot gaps and smaller TBs. The current logs do not retain a repeated gap-frequency or TB-size distribution, so this evidence is qualitative. It *narrows the likely cause* to split-path scheduling/polling/timing behavior but does not distinguish F1 arrival jitter from internal OAI producer–consumer cadence or establish a software root cause.

E. Airborne Payload, PWS over F1, and Service Validation

The performance numbers matter because they attach to a concrete payload. For the validated Jetson configuration, the board, B210 with access antennas, and RM500Q-GL modem kit total 657.4 g and draw approximately 28 W under sustained traffic. Adding a 100-g integration allowance gives 757.4 g. We

TABLE VI
MEASURED UAV-ORIENTED ELECTRONICS PAYLOAD

Item / metric	Measured value
Jetson Orin Nano board	151.4 g
USRP B210 + access antennas	217.3 g
RM500Q-GL modem kit + antennas	288.7 g
Core electronics mass	657.4 g
Integration allowance (+100 g)	757.4 g
Sustained-traffic electrical load	~28 W

do not convert 28 W into flight time because battery voltage, propulsion load, regulator efficiency, cooling, and airframe are aircraft-specific; the measurement is instead the electrical budget that a flight designer must reserve for the cellular payload.

Public-warning delivery is a distinct software contribution and a direct emergency-response requirement. A temporary cell deployed above a disaster zone should be able to broadcast safety instructions as soon as coverage is available; PWS is well suited to this role because the warning is cell-broadcast rather than individually addressed, so the network does not need the affected users' phone numbers or a destination list. In prior monolithic OAI work, SIB8 terminates inside one gNB process. In our split gNB, however, the warning originates at the ground CU while the remote DU owns MAC scheduling and the radio. Our patch therefore adds the missing F1AP Write-Replace Warning path: the CU packages the PWS system-information container and sends the request over SCTP/F1-C; the DU decodes it and hands SIB8 to MAC for SI scheduling [3]. In the validated single-segment case, the warning traverses CU→F1→DU→SIB8 and is displayed by the commercial handset. The available instrumentation records a 0.13-s CU-to-DU network-side event and about 0.8 s handset-visible latency; because these timestamps are not generated from one synchronized clock, we treat them as service-validation intervals, not a stage-by-stage latency decomposition.

As a secondary replication check, two DUs remain simultaneously active behind one CU under sustained traffic, retaining 91% and 93% of their individual single-DU baselines. This is a first-step shared-CU feasibility result rather than a scaling study, but together with PWS it shows that the lightweight remote-cell unit preserves both user data and application-visible broadcast service.

A synchronized Pi 5 sample further uses about 1.8 CPU cores and 1.4 GB RAM at 64.8 °C without late/UHD overflow markers after thread pinning. The B210 remains the validated access radio; a future B200mini-class radio could reduce payload mass further, but would require a new integrated power/thermal measurement [17].

V. DISCUSSION AND DESIGN IMPLICATIONS

The measurements give a direct answer to the UAV design question. Keeping the 5GC/CU on the ground avoids repeating shared compute on every aircraft; the remaining choice is how much performance to buy with the computer that does fly. In our fixed 106-PRB testbed, Jetson loses only 10–12.5% of

x86 DL across the three bearers, while Pi 5 accepts a larger performance reduction. Combined with the measured 657.4-g / 28-W Jetson-based electronics point, this makes the Jetson configuration the most attractive *performance-preserving* embedded option we tested. This is a testbed-specific design conclusion, not a claim that one Arm board is universally optimal.

Untethering the F1 link creates a second, largely separable cost. The 5G/WireGuard path preserves about 76–77% of each host's Ethernet DL, whereas the tested Wi-Fi/GRE route is lower and more variable. For an aerial system, this means compute and backhaul should be budgeted independently: selecting a faster DU does not remove the penalty of the wireless F1 path, and a good backhaul cannot compensate for an underpowered DU.

The BLER experiment adds an operational lesson that matters outside the headline throughput plot. A cell can have F1 established, a registered UE, and working IP traffic while link adaptation has collapsed to MCS 3. The controlled threshold intervention recovers $4.25\times$ DL without changing RF, UE, BWP, TDD, or F1. For a remote or airborne cell, such a failure would be particularly costly because the system would consume flight energy while providing a fraction of its expected capacity. The layered service gates and telemetry are therefore part of the deployment design, not merely debugging convenience.

The remaining x86 monolithic-split gap is separate from that BLER failure. Both states use MCS 26 with BLER 0.27/0.28; split Ethernet carries 104.8 Mb/s F1-U at 99.4 Mb/s application throughput, with zero observed F1-U loss/persistent queue, a 62% busiest DU real-time thread, and clean UHD counters. These observations make sustained RF, F1-capacity, queue, or gross CPU saturation poor explanations. The distinguishing trace is intermittent PRB/TB scheduling in split mode versus continuous full-PRB use monolithically. We therefore narrow the likely cause to split-path scheduling/polling/timing behavior. Quantitative slot-aligned timestamps for F1-U arrival, DU wake-up, grant construction, TB size, PRB occupancy, and HARQ feedback are still required to distinguish network arrival jitter from an internal OAI producer-consumer cadence.

The study intentionally stops before claiming flight endurance or universal bearer rankings. We test one n78/106-PRB profile, one commercial UE, and specific Wi-Fi and donor-cell routes. Likewise, the diagnostic BLER window was chosen after observing the failure state; it proves the controller mechanism in this setup, not a universal optimum. These boundaries do not weaken the central result: they define a reproducible laboratory baseline from which an airframe designer can trade DU performance, backhaul choice, payload mass, and power without first rebuilding the cellular stack from scratch.

VI. REPRODUCIBILITY ARTIFACT

To make the UAV-oriented baseline reproducible outside our lab, the public artifact pins the OAI revision, the published

PWS-over-F1 and instrumentation patches, sanitized scenario templates, host discovery, MTU/routing setup, Jetson SCTP kernel build, packet-path checks, service gates, and rollback workflow [14], [3], [15]. It also records the operational details that are brittle in a real lab: live B210 ownership, the RM500Q-GL donor/access boundary, Ethernet MTU restoration, GRE/WireGuard route selection, and detection of stale F1 traffic on management interfaces. Reproduction requires F1 on the intended interface, UE attachment to the access B210, a working PDU session, and clean return to the Ethernet baseline.

This artifact-first contract is important because several failure modes in the paper leave processes alive: wrong-interface F1 can still establish SCTP, and the default BLER state still registers the UE and carries IP traffic. The reproduction target is therefore the layered observable signature (configuration, packet path, radio state, service, and performance), not a successful launch command. Credentials and subscriber secrets are excluded.

VII. CONCLUSION

We built the split-DU testbed around a simple aerial design objective: keep shared 5GC/CU compute on the ground and make the repeated airborne unit as small and reproducible as possible without giving up useful real-radio performance. With commodity hardware, Jetson retains 88.4% of x86 Ethernet DL and 87.5–89.0% across all tested F1 bearers, while the validated Jetson/B210/RM500Q-GL electronics weigh 657.4 g and draw about 28 W. Across 400 repeated measurements, the resulting arithmetic means provide practical design baselines. A controlled BLER intervention recovers a $4.25\times$ hidden throughput collapse, and synchronized mono/split telemetry narrows the remaining split penalty to scheduling/timing behavior rather than a simple RF, F1-capacity, or CPU bottleneck. We also contribute, to the best of our knowledge, the first public OAI PWS-over-F1 path: a released F1AP Write-Replace Warning patch carries the warning from the ground CU to the remote DU, which broadcasts SIB8 to compatible handsets without per-user phone-number addressing. This connects the systems objective to the emergency use case directly: a lightweight cell flown over an affected area can provide an immediate area-wide warning service as well as user data. Two-DU operation further verifies first-step replication behind the shared ground CU. The result is not only an OAI setup that runs on Arm, but a reproducible remote-cell building block for future UAV experiments where every carried gram and watt matters.

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